

NMTC Problem-Set 3: Solutions

Instructions: The following are the solutions for Problem Set 3. The questions haven't been included this time so it's recommended to read this document along with the problem set document. The sources to the problems can be found on the last page of the document.

Problem 1: Some Functional Equations:

Proof. Since these are equations in $\mathbb{N}$, we will almost assuredly have to use induction.

1. We begin by letting $f(1) = c$. Plugging in $m = n = 1$ gives

$$f(2) = 2c + 2 = 2(c + 1)$$

Plugging in $m = 1, n = 2$ gives

$$f(3) = f(1) + f(2) + 4 \iff$$

$$f(3) = c + 2c + 2 + 4 \iff$$

$$f(3) = 3(c + 2)$$

We can prove easily, by induction(I won't do this but you will need to on the test or olympiad), that in general we have:

$$f(m) = m(c + m - 1)$$

for all natural numbers m . From the given information, we know that this must be the square of an integer, for all choices of m . We thus choose $m = p$, where p is an arbitrary prime. (This is a natural choice since we have very little information to work with, and primes have nice properties with which we could derive some constraints). We then have:

$$f(p) = p(c + p - 1) = k^2$$

where k is an integer. It is easy to see that p divides k^2 , and since p is a prime, it follows that p divides k . We write $k = tp$ and get:

$$p(c + p - 1) = t^2 p^2 \iff$$

$$c + p - 1 = t^2 p \iff$$

$$c - 1 = p(t^2 - 1)$$

This implies p divides $c - 1$, for all choices of a prime p . Hence, $c - 1 = 0$, so $c = 1$. Plugging this into the general form for $f(m)$ we derived above, we get: $f(m) = m^2$. It must be checked that this function works, and indeed it does, and this is the only solution.

2. With very little information about the function, it's natural to check for injectivity and surjectivity. The smallest value on the *RHS* is 2, and all values greater can be achieved, so the function is surjective on the set $\mathbb{N} \setminus \{1\}$.

To check for injectivity, we suppose $f(a) = f(b)$. Plugging into the equation we get:

$$a + b = f(f(a) + f(b)) = f(f(a) + f(a)) = 2a \implies a = b$$

Hence the function is injective on the whole domain.

We'd like to use the fact that the function is injective, so we want to find two pairs of values (a, b) , that give the same output. We see easily that

$$\begin{aligned} f(f(2) + f(2)) &= 2 + 2 = 4 = 1 + 3 = f(f(1) + f(3)) \implies \\ f(2) + f(2) &= f(1) + f(3) \end{aligned}$$

Since $f(2)$ is the average of $f(1)$ and $f(3)$, we have 2 cases. Either $f(1) < f(2) < f(3)$, OR $f(1) > f(2) > f(3)$. We consider these cases separately.

- (a) Case 1: $f(1) > f(2) > f(3)$

We derive a contradiction in this case, once again by induction. Suppose we have, for some k , that

$$f(1) > f(2) > \dots > f(k)$$

We then have

$$\begin{aligned} k + 2 &= f(f(k+1) + f(1)) = f(f(k) + f(2)) \implies \\ f(k+1) + f(1) &= f(k) + f(2) \end{aligned}$$

Since $f(1) > f(2)$, we see that $f(k) > f(k+1)$. This gives:

$$f(1) > f(2) > \dots$$

This is an infinitely decreasing sequence of natural numbers, which is absurd (You could say it contradicts the principle of infinite descent), and so we discard this case.

(b) Case 2: $f(1) < f(2) < f(3)$

Similar to the case above, we can use induction to prove that:

$$f(1) < f(2) < \dots$$

Since f is bijective on $\mathbb{N}_{\geq 2}$, we must have either $f(1) = 2$, or $f(1) = 1$. (Why?)

If $f(1) = 2$, then $f(2) = 3$, and so on, so we have $f(n) = n + 1$. By plugging this into the original functional equation, we see that we get $m + n + 3 = m + n$, which does not work.

Hence the only possibility is $f(1) = 1$, and so $f(m) = m$. It must be checked that this function works, and indeed it does. Hence this is the only solution, and we are done.

□

Problem 2: Combinatorics

Proof. 1. This is a problem for which we need to find an invariant/monovariant.

We know intuitively, that the numbers get closer and closer together, so the distance between the smallest and largest number is always non-increasing. Inspired by this, we define S to be the sum of the squares of all the numbers on the board at any step.

We look at what happens to S after a step.

We see that we have, taking $a - b \geq 2$

$$\begin{aligned} S &\rightarrow S - a^2 - b^2 + (a-1)^2 + (b+1)^2 \implies \\ S &\rightarrow S - 2(a-1-b) \end{aligned}$$

Since $a - b \geq 2$, $a - b - 1 \geq 1$, so S decreases by at least 2 after every step. However, S is also the sum of squares, so it can never be negative, meaning it cannot decrease indefinitely. Hence, we can only make a finite number of steps.

2. Notice that every time an animal eats another, the parity of the number of every animal changes by 1. For example, if a wolf eats a cat, we get 54 wolves, 16 cats, and 7 sparrows.

Thus, if two animals have different parities in the start, they must have different parities at the end. This tells us that the wolves and sparrows cannot both vanish, and the cats and sparrows cannot both vanish. It follows that the wolves and cats both vanish, and the last species left are the sparrows. Since all 55 wolves must vanish, we must have at least 55 different "eatings". Since every eating reduces the number of animals by 1, after 55 eatings we have at most $55+17+6-55=23$ animals remaining.

We now show that a sequence of eatings which results in 23 animals remaining is possible. To maximize the number of sparrows, the wolves first eat all the cats. This results in $17+6=23$ sparrows, and 38 wolves. Then, 19 wolves eat 19 sparrows and we are left with:

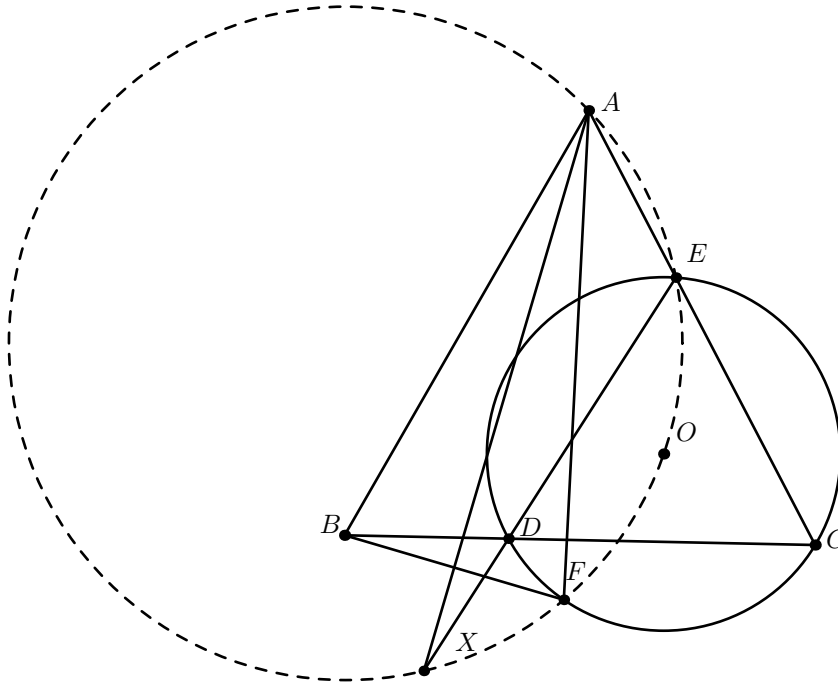
19 wolves, 19 cats, 4 sparrows.

The wolves then eat all the cats and go extinct themselves, so we are left with 23 sparrows at the end.

□

Problem 3: Cyclic Quads!

Solution.



Refer to the above diagram for the solution.

- (a) There are two cases to consider. Firstly, If E lies on the segment AC , we have that:

$$\begin{aligned}
 \angle ACF &= \angle ECF \\
 &= \frac{\angle EOF}{2} = \frac{180^\circ - \angle EAF}{2} \\
 &= \frac{180^\circ - \angle CAF}{2} \\
 &= \frac{180^\circ - (180^\circ - \angle AFC - \angle ACF)}{2} \\
 &= \frac{\angle AFC + \angle ACF}{2}.
 \end{aligned}$$

Re-arranging the above relation gives us that $\angle ACF = \angle AFC$ from which we deduce $AF = AC = AB$, as needed.

Our second case is if E lies on the extension of line AC past A (Not shown on a diagram due to laziness), then we have that:

$$\begin{aligned}\angle ACF &= \angle ECF \\ &= \frac{\angle EOF}{2} = \frac{\angle EAF}{2} \\ &= \frac{180^\circ - \angle FAC}{2} \\ &= \frac{180^\circ - (180^\circ - \angle AFC - \angle ACF)}{2} \\ &= \frac{\angle AFC + \angle ACF}{2}.\end{aligned}$$

Re-arranging the above relation gives us that $\angle ACF = \angle AFC$ from which we deduce $AF = AC = AB$, as needed.

- (b) Observe that by part (a), we have that A is the circumcentre of $\triangle FBC$. Hence, we have:

$$\begin{aligned}\angle XAF &= \frac{1}{2}\angle BAF \\ &= \angle BCF = \angle DCF \\ &= \angle DEF = \angle XEF\end{aligned}$$

So, $\angle XAF = \angle XEF$. This implies that X, F, E, A lie on a circle. But A, E, O, F also lie on a circle. Hence, X, F, O, E, A lie on a circle, as needed.

- (c) Observe that:

$$\begin{aligned}\angle XFO &= 180^\circ - \angle XEO \\ &= 180^\circ - \angle DEO \\ &= 180^\circ - \angle EDO \\ &= \angle XDO\end{aligned}$$

Since $\angle OFD = \angle ODF$. This implies that $\angle XFD = \angle XDF$. Hence, $XF = XD$, but $XF = XB$ (Why?) as well so we're done.

Problem 4: Integer Polynomials

Solution.

Proof. (a) Let $P(n) = x_k n^k + x_{k-1} n^{k-1} + \cdots + x_1 n + x_0$. We have that:

$$\begin{aligned} P(a) - P(b) &= (x_k a^k + \cdots + x_1 a + x_0) - (x_k b^k + \cdots + x_1 b + x_0) \\ &= x_k(a^k - b^k) + x_{k-1}(a^{k-1} - b^{k-1}) + \cdots + x_1(a - b) \end{aligned}$$

It follows that $a - b$ divides each term in the above expressive. Hence, $a - b \mid P(a) - P(b)$ indeed holds.

- (b) Keeping part (a) in mind, if we can find infinitely many x such that $m \mid 2^{kx} - 2^k$, we'll be done. This is equivalent to finding x such that:

$$\begin{aligned} 2^{kx} &\equiv 2^k \pmod{m} \\ \iff 2^{x-1} &\equiv 1 \pmod{m} \end{aligned}$$

This reminds us of Euler's generalization of Fermat's Little Theorem! Indeed, if we let $x = l\phi(m) + 1$ for any integer l , where $\phi(m)$ denotes Euler's totient function. Then the above equation holds, and we're done.

- (c) Take any prime p . Note that $p \mid kp \mid P(2^{kp})$ for any integer k . By Fermat's little theorem, we also have:

$$0 \equiv 2^{kp} \equiv 2^k \pmod{p} \implies$$

$$0 \equiv P(2^{kp}) \equiv P(2^k) \pmod{p}$$

Hence, $p \mid P(2^k)$. But since p can be any prime p , this implies that a prime greater than $P(2^k)$ divides $P(2^k)$. Hence, $P(2^k) = 0$ must hold. Since k can be any integer, this means that $P(n)$ has infinitely many roots. This is only possible if $P(n) \equiv 0$, which is our only solution. $\square$

Sources:

1. Functional Equations from imomath.com
2. 2025 Pakistan IMO Second Training Camp practice problems
3. 2022 Japanese Mathematical Olympiad Final Round P3
4. (a) Well known
(b) Well known
(c) 2016 ELMO P4